

# NASSAU CANDY DISTRIBUTOR – FACTORY OPTIMIZATION ANALYSIS

## Project Submission Report

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**Internship Program:** Unified Mentor

**Domain:** Data Analytics

Tools Used: Python, Jupyter Notebook, Power BI

### 1. Introduction

Efficient factory allocation and distribution management are critical components of supply chain optimization.

Nassau Candy Distributor manages product distribution across multiple factories and regional markets. Effective factory-product assignment directly impacts shipping performance, operational cost, customer satisfaction, and profitability.

This project focuses on analyzing logistics performance, delivery efficiency, and product distribution trends to support data-driven factory optimization decisions.

### 2. Problem Statement

Nassau Candy Distributor currently relies on static product-factory allocation processes.

This creates several operational challenges:

- Inefficient shipping routes
- High delivery lead times
- Increased logistics costs
- Uneven regional performance
- Limited decision-making visibility

Without structured analytical insights, optimizing distribution performance becomes difficult.

### 3. Project Objectives

The primary objectives of this project are:

- Analyze regional delivery performance
- Evaluate shipping mode utilization
- Measure product-level sales performance
- Identify operational bottlenecks
- Generate logistics optimization insights
- Support better factory allocation decisions

### 4. Methodology

The project was completed using Python for analytical processing and Power BI for dashboard visualization.

Step 1: Data Loading

The dataset was imported into Python using Pandas.

Step 2: Data Cleaning

Data preprocessing included:

- Handling missing values
- Standardizing column names
- Data consistency checks
- Data type validation

Step 3: Exploratory Data Analysis

Performed analysis on:

- Regional orders
- Delivery trends
- Shipping modes
- Product performance
- Gross profit distribution

Step 4: Visualization

Interactive dashboard built using Power BI.

## **6. Python Analysis**

Python analysis was conducted using:

- Pandas
- NumPy
- Matplotlib

### **Key Analysis Performed**

#### **Regional Delivery Analysis**

Compared delivery performance across regions.

Monthly Order Trend Analysis

Observed order patterns over time.

Shipping Mode Distribution

Analyzed frequency of shipping methods.

Product Performance Analysis

Measured product-level sales and profitability.

Gross Profit Analysis

Compared financial performance across products.

## **7. Dashboard Insights**

The Power BI dashboard provides interactive logistics performance monitoring.

Dashboard Components

- KPI Cards
- Total Orders
- Total Sales
- Average Delivery Days
- Total Gross Profit

## Regional Delivery Chart

Displays delivery performance by region.

## Monthly Order Trend

Tracks order fluctuations over time.

## Shipping Mode Distribution

Shows shipping method usage.

## Product Performance Table

Displays product-level performance metrics.

## Interactive Region Filter

Enables region-based analysis.

## 8. Key Findings

The analysis revealed several important insights.

### 1. Regional Performance Variation

Delivery performance differs significantly across regions.

### 2. Shipping Mode Dependence

Standard Class dominates shipping operations.

### 3. Revenue Concentration

A limited set of products contributes significantly to sales...

### 4. Delivery Optimization Opportunity

Certain regional routes show potential for improvement.

### 5. Profitability Differences

High sales do not always correspond to high profitability.

## 9. Recommendations

Based on the analysis, the following recommendations are proposed.

### Optimize Factory Allocation

Reassign products to more efficient factories.

### Improve Delivery Planning

Focus operational improvements on slower-performing regions.

### Review Shipping Strategy

Balance delivery speed and operational cost.

### Prioritize High-Profit Products

Increase operational focus on profitable product lines.

### Continuous Performance Monitoring

Use dashboard insights for regular logistics review.

## 10. Conclusion

This project successfully transformed operational distribution data into actionable business intelligence.

Using Python analysis and Power BI dashboard visualization, valuable insights were generated regarding regional delivery efficiency, product performance, shipping behavior, and profitability.

The findings support data-driven factory optimization decisions that can improve operational efficiency, reduce logistics costs, and enhance customer satisfaction.

This analytical framework enables Nassau Candy Distributor to make smarter logistics decisions.

### Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Jupyter Notebook
- Power BI

Microsoft Word

### Future Scope

- Future enhancements may include:
- Predictive delivery modeling
- Machine learning-based factory recommendation
- Real-time logistics dashboard integration
- Automated route optimization.